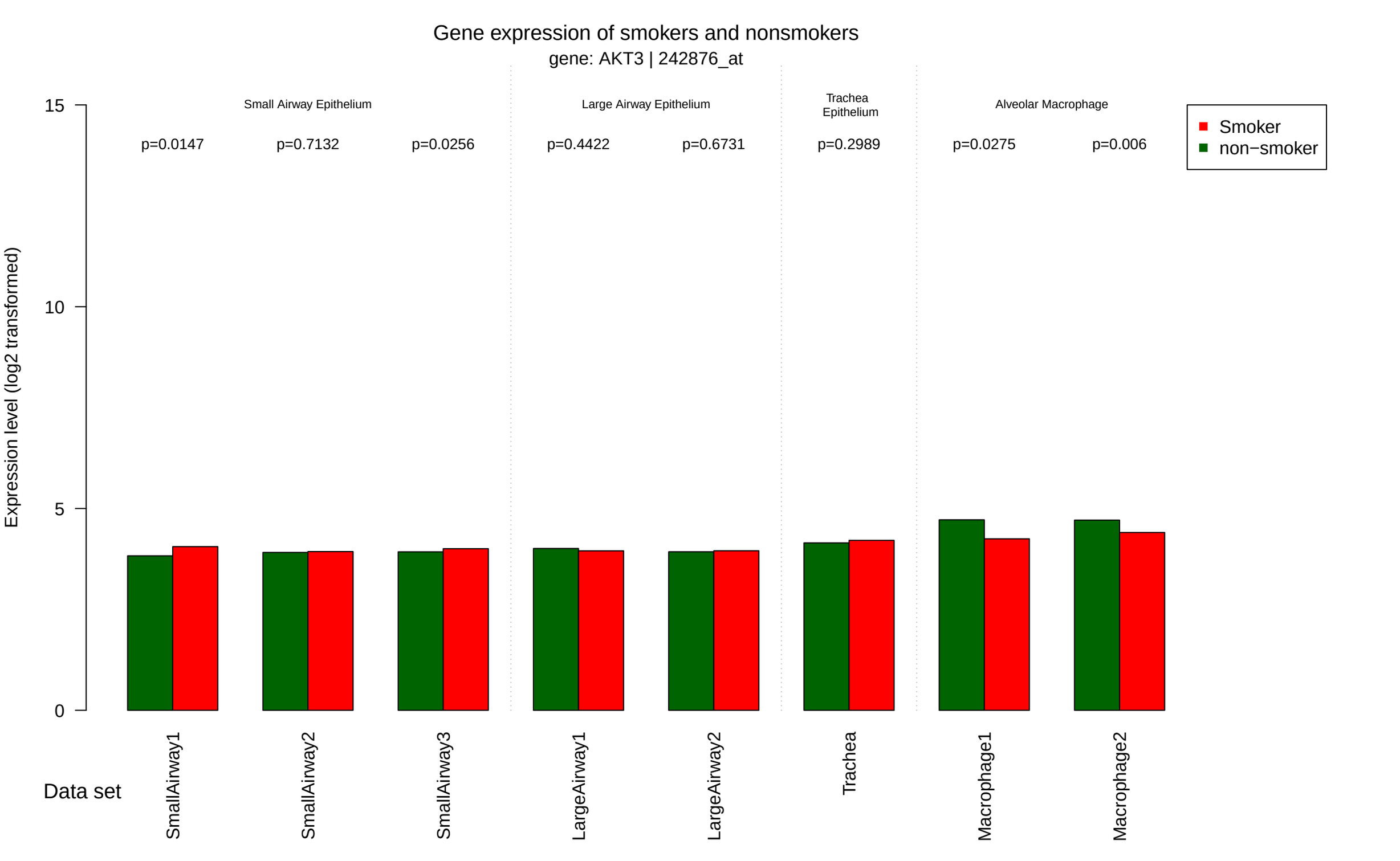
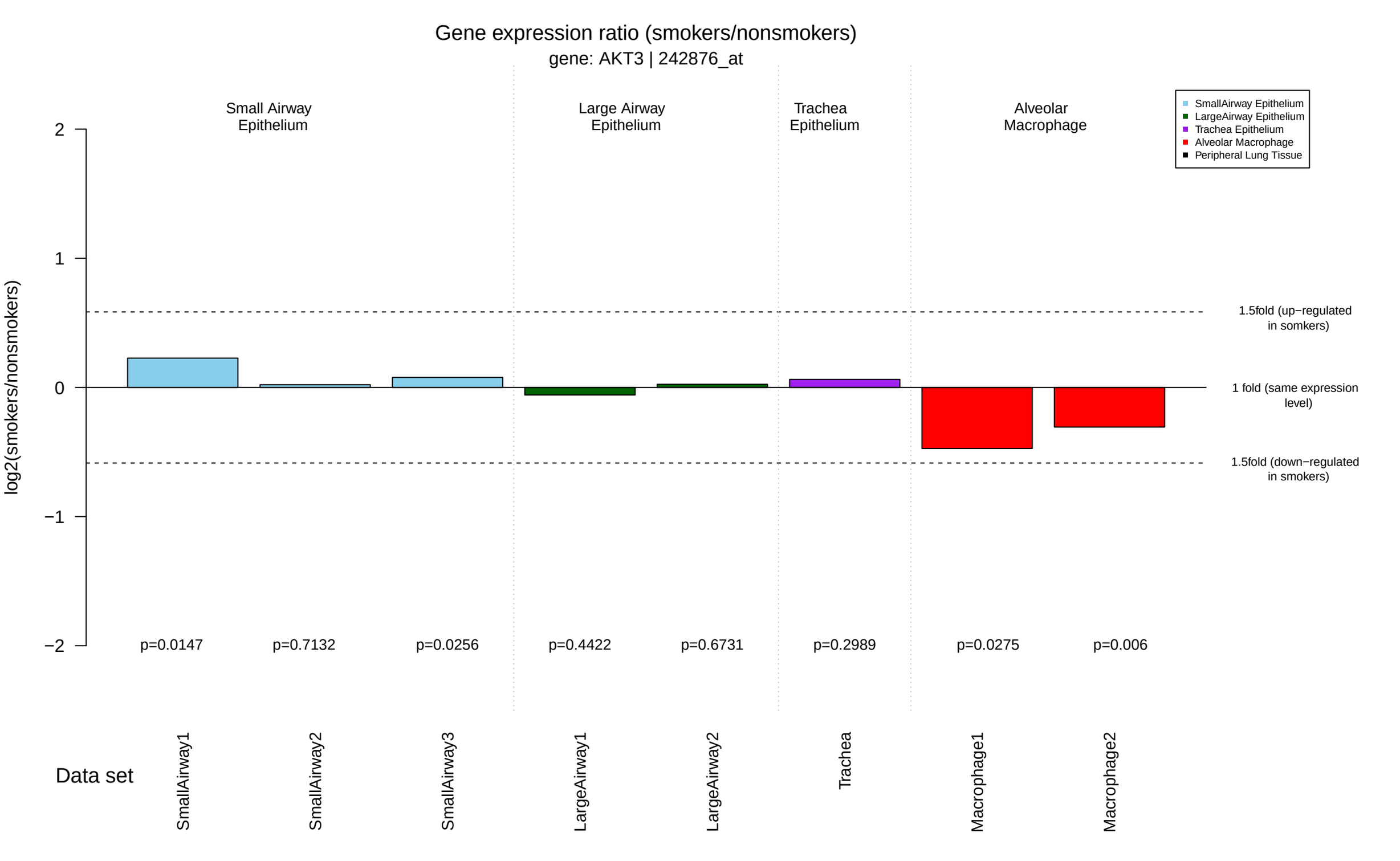
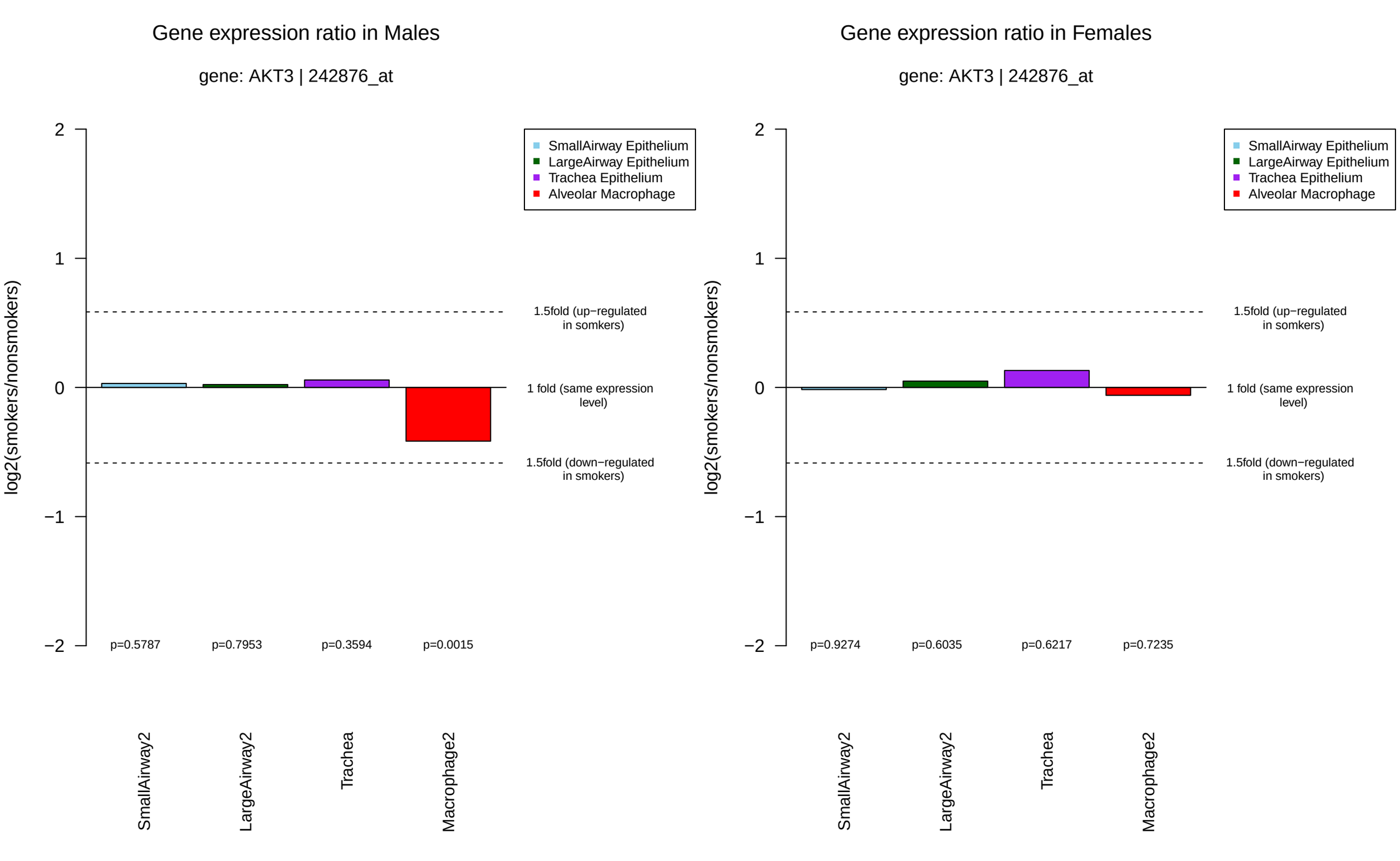
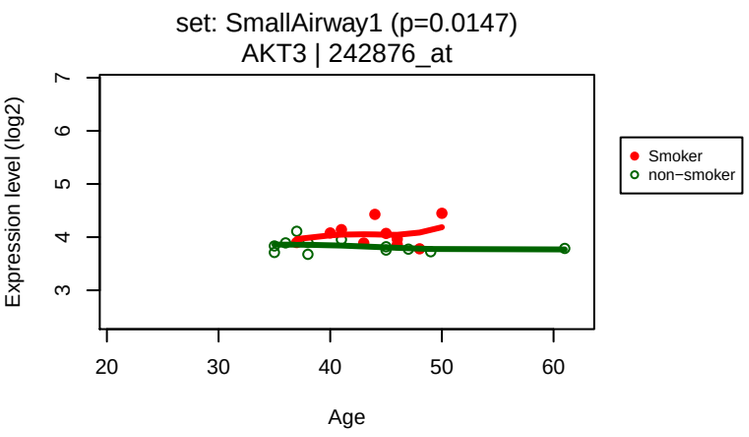
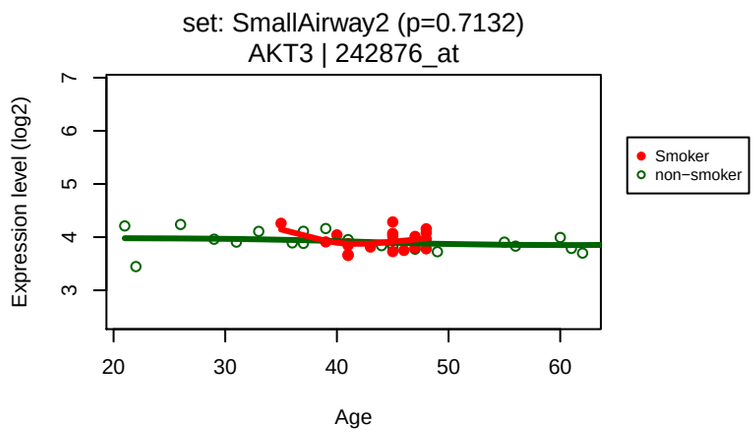
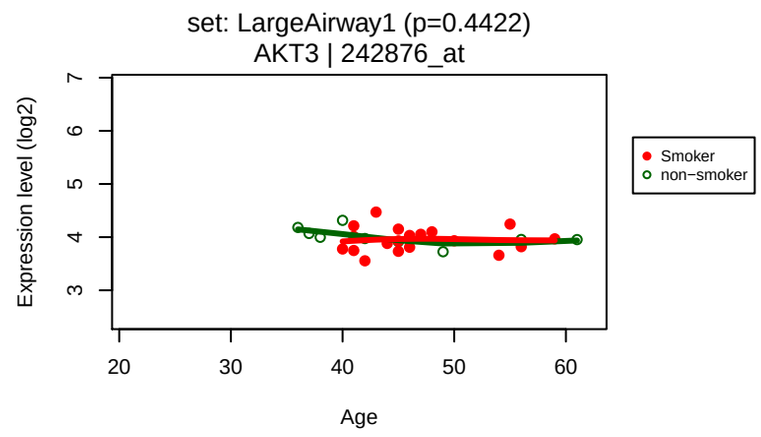

Gene expression of smokers and nonsmokers
gene: AKT3 | 242876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0147
p=0.7132
p=0.0256
p=0.4422
p=0.6731
p=0.2989
p=0.0275
p=0.006
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

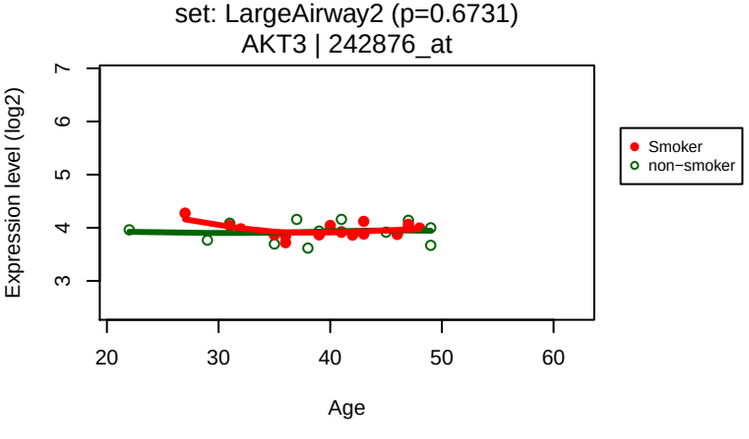

Gene expression ratio (smokers/nonsmokers)
gene: AKT3 | 242876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0147
p=0.7132
p=0.0256
p=0.4422
p=0.6731
p=0.2989
p=0.0275
p=0.006
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

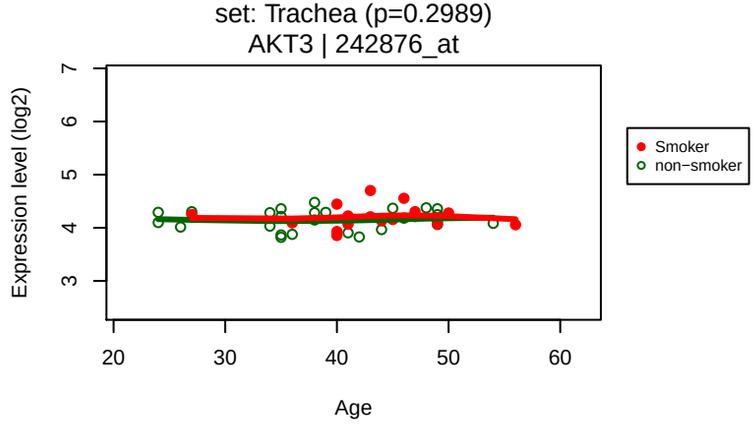

Gene expression ratio in Males
gene: AKT3 | 242876_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5787
p=0.7953
p=0.3594
p=0.0015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT3 | 242876_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9274
p=0.6035
p=0.6217
p=0.7235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

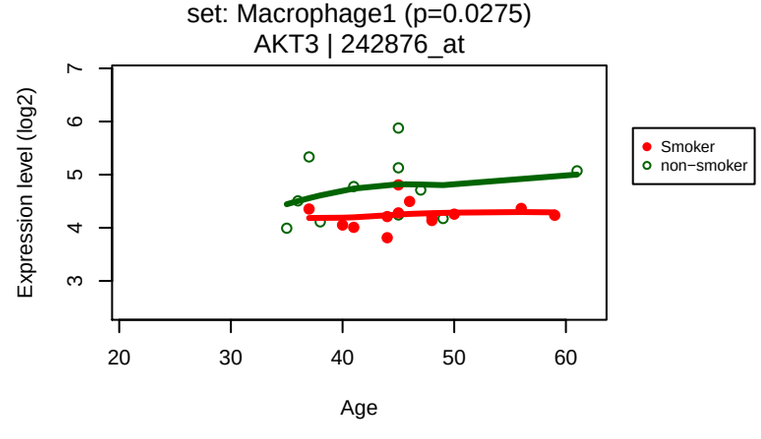

set: SmallAirway1 (p=0.0147)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker

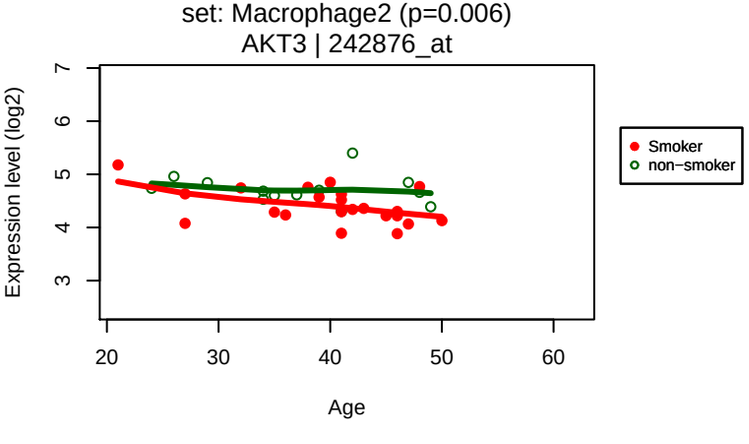

set: SmallAirway2 (p=0.7132)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker

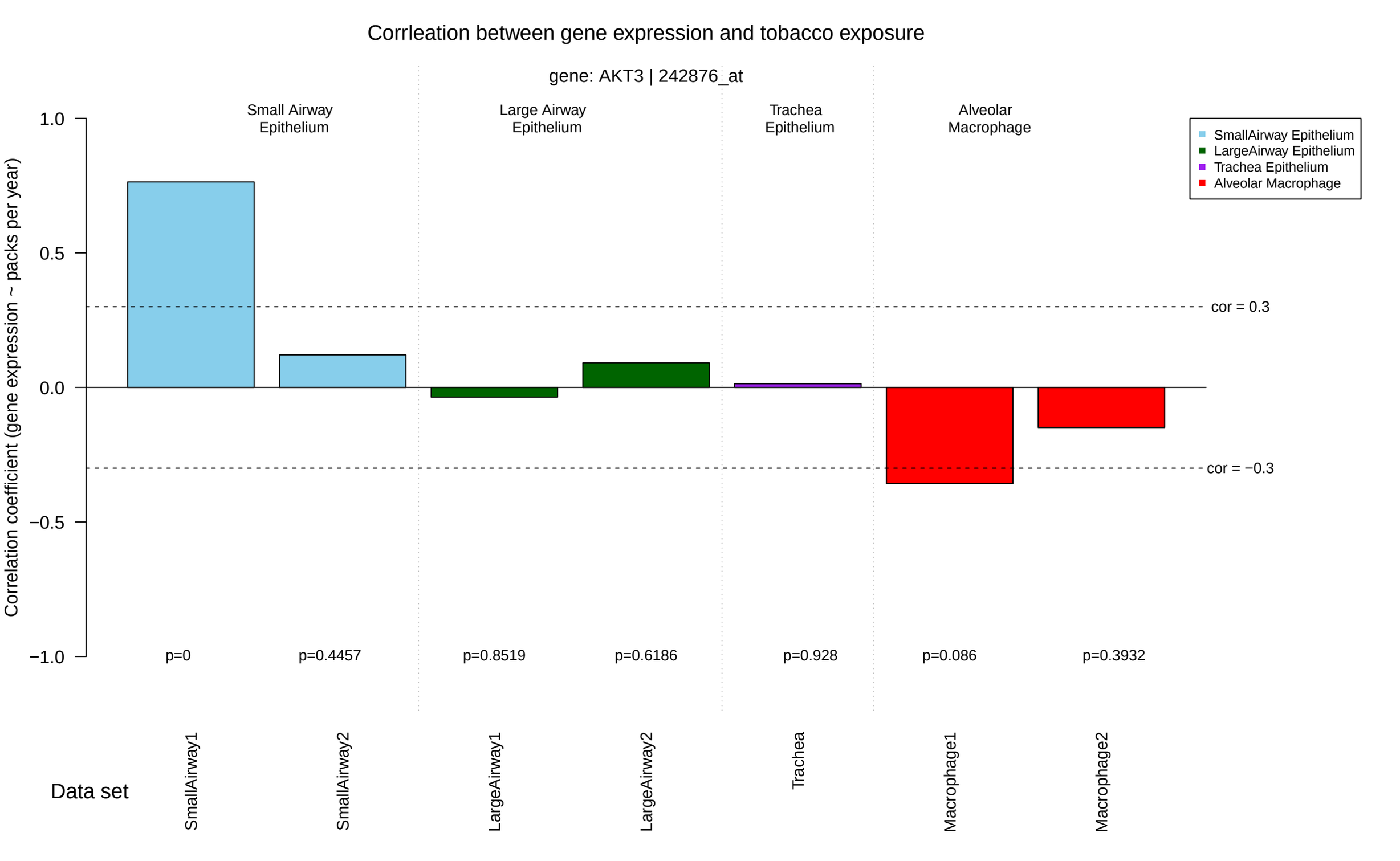

set: LargeAirway1 (p=0.4422)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6731)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2989)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0275)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.006)
AKT3 | 242876_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT3 | 242876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0
p=0.4457
p=0.8519
p=0.6186
p=0.928
p=0.086
p=0.3932
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2